

Tristan Misko

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SUMMARY

Full-stack software engineer who took a research team off an error-prone manual spreadsheet process and onto a production web app and automated pipelines — as sole engineer, owning architecture through deployment and on-call. My work centers on delivering systems that automate digital drudgery, decrease friction, and increase the leverage of teams and organizations, from that platform to a CLI tool that cut footnote-checking time for antitrust consultants by more than half.

SKILLS

Full-Stack Software Engineering	Go; Python; Flask; Click; MongoDB; TypeScript; JavaScript; Lua; HTML; CSS; Git; GitHub Actions; CI/CD
Data Engineering	Data Pipelines; ETL; Database Management; REST APIs; Web Scraping
Infrastructure & DevOps	Linux; Bash; Apache; NGINX; Linode; SSH; Server Administration

EXPERIENCE

Institute for Research on Labor and Employment

Aug 2023 – Present

Software & Data Engineer | Berkeley, CA

- Built and shipped a full-stack research platform (Flask, MongoDB, JavaScript) as sole engineer, replacing an error-prone manual Google Sheets process with automated pipelines across 60+ sources
- Reframed data collection as a programmatic problem — pulling one country reproducibly costs nearly the same as pulling all available — growing coverage from 49 to 100+ countries (67 at >80%) and temporal range from 2018 to 2000-2023
- Designed a five-stage DAG pipeline (collect, clean, compute, score, finalize) with fail-fast validation over several million records, scoring via numpy matrix ops for sub-second recomputation
- Owned production ops: root-caused MongoDB OOM kills, migrated Apache to NGINX off the thread-per-worker model, and ship zero-downtime semver-tagged releases via GitHub Actions over SSH at 95%+ uptime, while leading 3 teams (20 undergraduates)

Berkeley Research Group, LLC

Aug 2023 – Jul 2024

Economics & Damages Associate | Emeryville, CA

- Built reproducible Stata MP ingest-parse-clean pipelines normalizing dozens of inconsistently-formatted Excel feeds into datasets up to 500GB — reproducible from raw inputs, as the do-files shipped as litigation backups that opposing counsel re-ran for errors
- Kept overnight VM jobs under memory and budget caps via selective column loads, type management, and chunking; ran coverage/sanity/outlier checks that caught real defects (e.g., a balance-sheet/income-statement miscomputation)

Institute for Research on Labor and Employment

Aug 2021 – Aug 2023

Research Data Analyst | Berkeley, CA

- Designed quantitative policy indicators and wrote regression analysis code in R and Python to evaluate policy-outcome relationships
- Led research teams of 6-8 undergraduates, managing data collection across 60+ sources using Google Sheets and manual workflows

Earlier: Institute for Research on Labor and Employment — Research Apprentice (Aug 2020 – Aug 2021)

SELECTED PROJECTS

Arachne

Apr 2026 – Present

- Architected a 13-crate Rust backend (axum, tokio, sqlx) around a parallel DAG execution engine with lock-free single-owner scheduling, deadlock-free resource semaphores, and lossless pause/resume across server restarts at human-in-the-loop gates
- Built a structurally-hermetic code sandbox (Starlark) guaranteeing pure, deterministic, side-effect-free execution of agent-authored transforms, hardened by a never-shrinking adversarial escape-test corpus
- Wrote a from-scratch React 18 node-graph canvas (no node-editor library) with a single-coordinate-space transform and viewport-culled DOM rendering, holding a Playwright-enforced p95 < 20ms pan/zoom budget
- Drove development with a multi-agent autonomous loop running unattended in Docker — task-graph with preconditions/definitions-of-done, worktree-per-agent isolation, a hard egress allowlist, and disk/hang watchdogs — completing 340+ tracked tasks

BinderBuilder

Feb 2024 – Apr 2024

- Built a Python CLI tool (Click, fzf, MuPDF) using NLP (tokenization, TF-IDF, fuzzy matching) to automate footnote verification for expert reports, auto-highlighting cited text in candidate matches
- Reduced footnote verification time from 60s to 25s per footnote, measured against timecard baselines, saving hours of billable time per report

Switchboard

Mar 2026 – May 2026

- Built a Go daemon that correlates each agent process across three namespaces with no shared ID (/proc, the terminal multiplexer, the Wayland compositor), anchoring the join on the kernel-owned controlling TTY
- Detected process death by polling each PID's pidfd rather than scanning /proc or trapping signals, observing exit the instant a process becomes a zombie

EDUCATION

University of California, Berkeley

Aug 2018 – May 2022

B.A. Applied Mathematics | B.A. Economics, Honors